Problem Set - 10

1 Design the following counters:

(a) A mod-7 asynchronous counter using J-K FFs

(b) A mod-12 asynchronous counter using J-K FFs

(c) A mod-9 synchronous counter using J-K FFs

(d) A mod-7 synchronous counter using S-R FFs

(e) A mod-12 synchronous counter using T FFs

(f) A mod-14 synchronous counter using D FFs

Solution:

(a) Design of a Mod-7 asynchronous counter using J-K flip-flops.

The truth table for Mod-7 counter is shown below. It has 7 states. After the 7th pulse it comes back to starting state, i.e., "000". It requires 3 flip-flops.

Truth table

After pulses	**State**	**R**
	Q_3 Q_2 Q_1	
0	000	0
1	001	0
2	010	0
3	011	0
4	100	0
5	101	0
6	110	0
7	111 ↓↓↓ 000	1
8	001	

$$\bar{R} = \overline{Q_3 Q_2 Q_1}$$

Connect all J's and K's to '1'. Connect Q_3, Q_2 and Q_1 to a 3 input NAND gate. The output of the NAND gate is connected to $\overline{CLR}$ terminals of all flip-flops.

(b) Design of a Mod-12 asynchronous counter using J-K flip-flops.

The truth table for Mod-12 counter is shown below. It has 12 states. After the 12th pulse it comes back to starting state, i.e., "0000". It requires 4 flip-flops.

Truth table

After pulses	**State**	**R**
	Q_4 Q_3 Q_2 Q_1	
0	0000	0
1	0001	0
2	0010	0
3	0011	0
4	0100	0
5	0101	0
6	0110	0
7	0111	0
8	1000	0
9	1001	0
10	1010	0
11	1011	0
12	1100 ↓↓↓↓ 0000	1
13	0001	

$$\bar{R} = \overline{Q_4 Q_2}$$

To realize the counter using J-K FFs connect all J's and K's to '1'. Connect Q_4 and Q_2 to a 2 input NAND gate. The output of the NAND gate is connected to $\overline{CLR}$ terminals of all flip-flops.

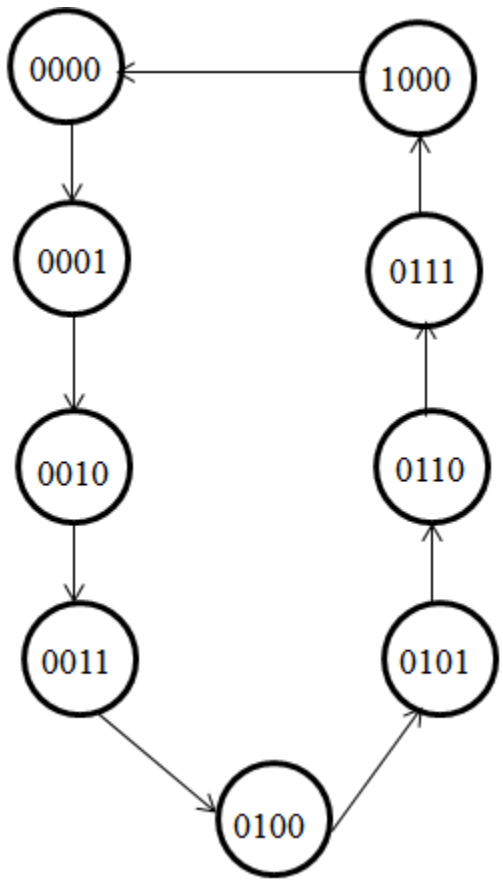
(c) Design of a Mod-9 synchronous counter using J-K flip-flops.

Step 1: The number of flip-flops

The counter has 9 states. $9 \leq 2^n$. So n = 4. It requires 4 flip-flops. States 1010, 1011, 1100, 1101, 1110 and 1111 are not valid. So they are don't cares.

Step 2: The state diagram

The state diagram of the counter is shown below.



State diagram

Step 3: The type of flip-flops and the excitation table

J-K flip-flops are selected and the excitation table of Mod-9 counter using J-K flip-flops is shown below.

Excitation table

PS	NS	Required Excitations

$Q_4\ Q_3\ Q_2\ Q_1$	$Q_4\ Q_3\ Q_2\ Q_1$	$J_4\ K_4$	$J_3\ K_3$	$J_2\ K_2$	$J_1\ K_1$
0 0 0 0	0 0 0 1	0 X	0 X	0 X	1 X
0 0 0 1	0 0 1 0	0 X	0 X	1 X	X 1
0 0 1 0	0 0 1 1	0 X	0 X	X 0	1 X
0 0 1 1	0 1 0 0	0 X	1 X	X 1	X 1
0 1 0 0	0 1 0 1	0 X	X 0	0 X	1 X
0 1 0 1	0 1 1 0	0 X	X 0	1 X	X 1
0 1 1 0	0 1 1 1	0 X	X 0	X 0	1 X
0 1 1 1	1 0 0 0	1 X	X 1	X 1	X 1
1 0 0 0	0 0 0 0	X 1	0 X	0 X	0 X

In Mod-9 counter states 1001, 1010, 1011, 1100, 1101, 1110 and 1111 do not occur. Hence they are don't cares, i.e., $\sum$ d (9, 10, 11, 12, 13, 14, 15).

Step 4: The minimal expressions for the excitation are obtained from the K-maps shown below.

Q_4Q_3 \ $Q_2\ Q_1$	00	01	11	10
00	0	1	3	2
01	4	5	7: 1	6
11	12: X	13: X	15: X	14: X
10	8: X	9: X	11: X	10: X

$J_4 = Q_3Q_2Q_1$

Q_4Q_3 \ $Q_2\ Q_1$	00	01	11	10
00	0	1	3: 1	2
01	4: X	5: X	7: X	6: X
11	12: X	13: X	15: X	14: X
10	8	9: X	11: X	10: X

$J_3 = Q_2Q_1$

From the table $K_4 = 1$ and $K_1 = 1$

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0: X	1: X	3: X	2: X
01	4:	5:	7: 1	6:
11	12: X	13: X	15: X	14: X
10	8: X	9: X	11: X	10: X

$K_3 = Q_2Q_1$

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0:	1: 1	3: X	2: X
01	4:	5: 1	7: X	6: X
11	12: X	13: X	15: X	14: X
10	8:	9: X	11: X	10: X

$J_2 = Q_1$

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0: X	1: X	3: 1	2:
01	4: X	5: X	7: 1	6:
11	12: X	13: X	15: X	14: X
10	8: X	9: X	11: X	10: X

$K_2 = Q_1$

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0: 1	1: X	3: X	2: 1
01	4: 1	5: X	7: X	6: 1
11	12: X	13: X	15: X	14: X
10	8:	9: X	11: X	10: X

$J_1 = \overline{Q_4}$

K-maps

The design equations for the mod-9 synchronous counter using J-K FFs are

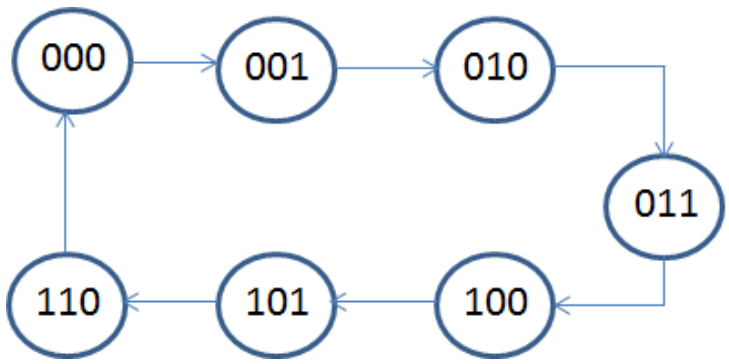
$$J_4 = Q_3Q_2Q_1, K_4 = 1, J_3 = Q_2Q_1, K_3 = Q_2Q_1, J_2 = Q_1, K_2 = Q_1, J_1 = \overline{Q_4}, K_1 = 1$$

A logic diagram can be drawn based on those equations.

(d) Mod-7 synchronous counter using S-R flip-flops.

Step 1: The Mod-7 counter has 7 states. It requires 3 flip-flops. State "111" is invalid. So it is a don't care.

Step 2: The state diagram of the counter is shown below.



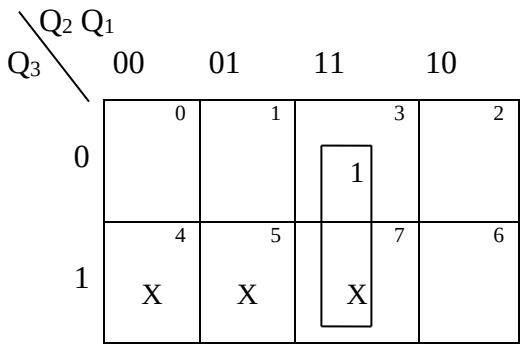
State diagram

Step 3: S-R flip-flops are selected and the excitation table of the Mod-7 counter using S-R flip-flops is shown below.

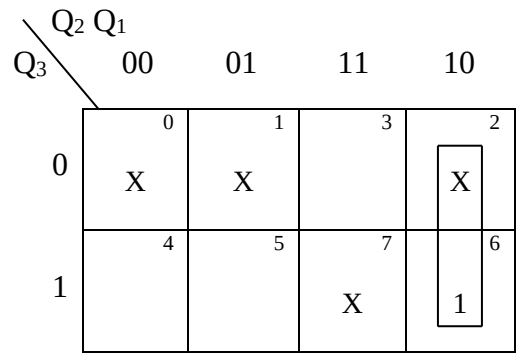
Excitation table

PS	**NS**	**Required Excitations**
Q_3 Q_2 Q_1	**Q_3 Q_2 Q_1**	**S_3 R_3 S_2 R_2 S_1 R_1**
0 0 0	0 0 1	0 X 0 X 1 0
0 0 1	0 1 0	0 X 1 0 0 1
0 1 0	0 1 1	0 X X 0 1 0
0 1 1	1 0 0	1 0 0 1 0 1
1 0 0	1 0 1	X 0 0 X 1 0
1 0 1	1 1 0	X 0 1 0 0 1
1 1 0	0 0 0	0 1 0 1 0 X

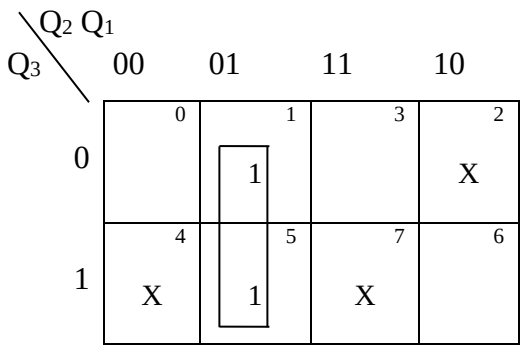
Step 4: The minimal expressions for excitations are obtained using K-maps as shown below.



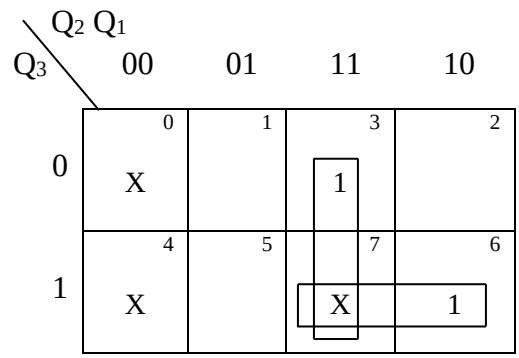
$S_3 = Q_2 Q_1$



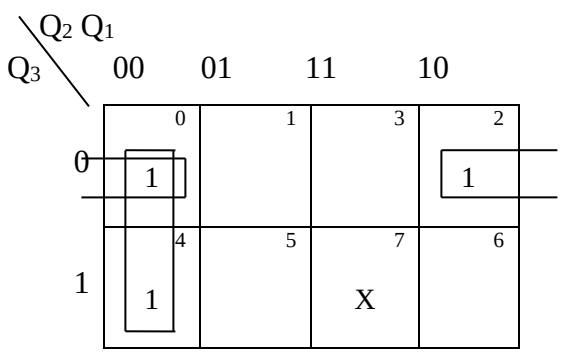
$R_3 = Q_2 \overline{Q_1}$



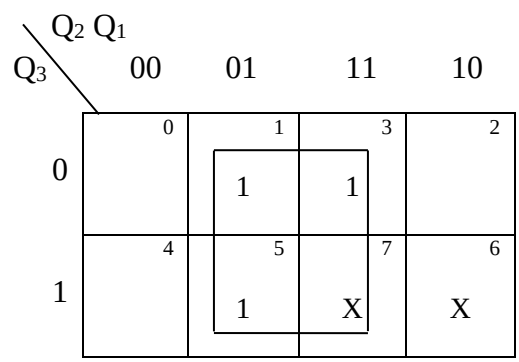
$S_2 = \overline{Q_2} Q_1$



$R_2 = Q_3 Q_2 + Q_2 Q_1$



$S_1 = \overline{Q_2}\,\overline{Q_1} + \overline{Q_3}\,\overline{Q_1}$



$R_1 = Q_1$

K-maps

So the design equations for the mod-7 synchronous counter using S-R FFs are $S_3 = Q_2Q_1$,

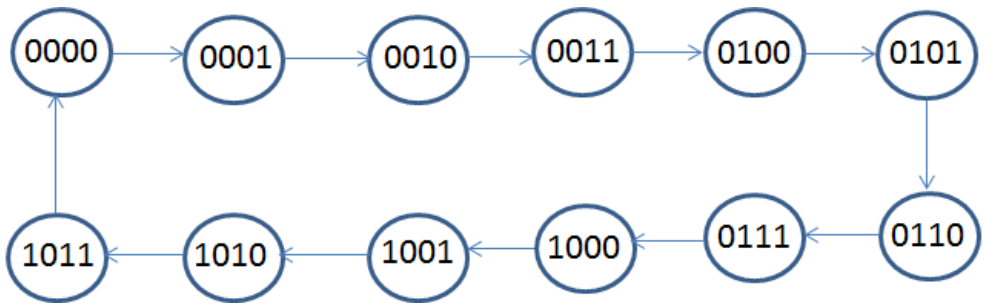
$R_3 = Q_2\overline{Q}_1$; $S_2 = \overline{Q_2}Q_1$; $R_2 = Q_3Q_2 + Q_2Q_1$; $S_1 = \overline{Q_2}\,\overline{Q_1} + \overline{Q_3}\,\overline{Q_1}$; $R_1 = Q_1$

A logic diagram can be drawn based on the above expressions

(e) Mod-12 synchronous counter using T flip-flops.

Step 1: The number of flip-flops required n is such that $12 < 2^n$. So n = 4. States 1100, 1101, 1110 and 1111 are invalid. So the corresponding minterms are don't cares.

Step 2: The state diagram of the counter is shown below.



State diagram

Step 3: The excitation table of mod-12 counter using T flip-flops is shown below.

Excitation table

PS	**NS**	**Required Excitations**
Q_4 Q_3 Q_2 Q_1	**Q_4 Q_3 Q_2 Q_1**	**T_4 T_3 T_2 T_1**
0 0 0 0	0 0 0 1	0 0 0 1
0 0 0 1	0 0 1 0	0 0 1 1
0 0 1 0	0 0 1 1	0 0 0 1
0 0 1 1	0 1 0 0	0 1 1 1
0 1 0 0	0 1 0 1	0 0 0 1
0 1 0 1	0 1 1 0	0 0 1 1
0 1 1 0	0 1 1 1	0 0 0 1
0 1 1 1	1 0 0 0	1 1 1 1
1 0 0 0	1 0 0 1	0 0 0 1

1 0 0 1	1 0 1 0	0 0 1 1
1 0 1 0	1 0 1 1	0 0 0 1
1 0 1 1	0 0 0 0	1 0 1 1

Step 4: The minimal expressions for excitations T_4, T_3, T_2, and T_1 are obtained using K-Maps as shown below.

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0	1	3	2
01	4	5	7: 1	6
11	12: X	13: X	15: X	14: X
10	8	9	11: X	10

$T_4 = Q_4 Q_2 Q_1 + Q_3 Q_2 Q_1$

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0	1	3: 1	2
01	4	5	7: 1	6
11	12: X	13: X	15: X	14: X
10	8	9	11	10

$T_3 = \overline{Q}_4 Q_2 Q_1$

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0	1: 1	3: 1	2
01	4	5: 1	7: 1	6
11	12: X	13: X	15: X	14: X
10	8	9: 1	11: 1	10

$T_2 = Q_1$

K-maps

From the excitation table observe that $T_1 = 1$

The design equations for the mod-12 synchronous counter using T FFs are

$$T_4 = Q_4Q_2Q_1 + Q_3Q_2Q_1$$

$$T_3 = \overline{Q_4}Q_2Q_1$$

$$T_2 = Q_1$$

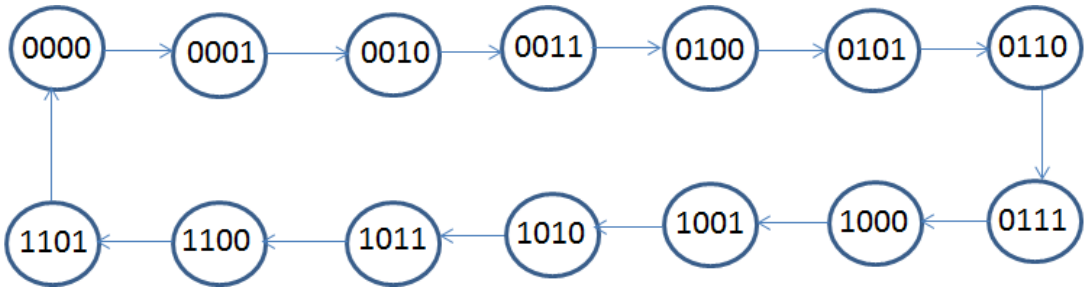
$$T_1 = 1$$

A logic diagram can be drawn based on these expressions.

(f) Design of a Mod-14 synchronous counter using D flip-flops.

Step 1: The number of flip-flops required n is such that $14 < 2^n$. So n = 4. States 1110 and 1111 are invalid. So the corresponding minterms are don't cares.

Step 2: The state diagram of the mod-14 counter is shown below.



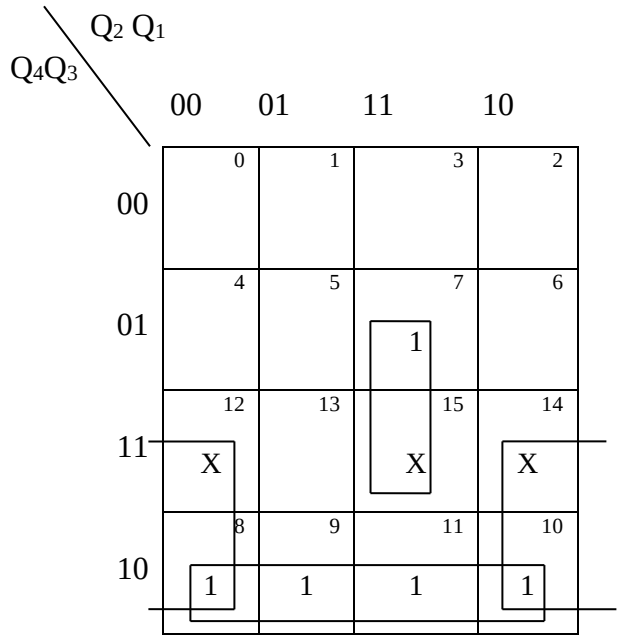
State diagram

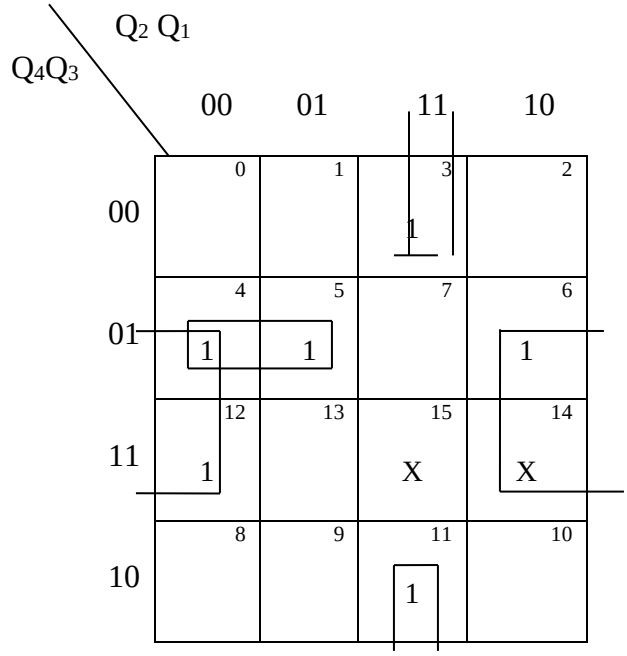
Step 3: The excitation table of the mod-14 counter using D flip-flops is shown below.

Excitation table

PS	**NS**	**Required Excitations**
Q_4 Q_3 Q_2 Q_1	**Q_4 Q_3 Q_2 Q_1**	**D_4 D_3 D_2 D_1**
0 0 0 0	0 0 0 1	0 0 0 1
0 0 0 1	0 0 1 0	0 0 1 0
0 0 1 0	0 0 1 1	0 0 1 1
0 0 1 1	0 1 0 0	0 1 0 0
0 1 0 0	0 1 0 1	0 1 0 1
0 1 0 1	0 1 1 0	0 1 1 0
0 1 1 0	0 1 1 1	0 1 1 1
0 1 1 1	1 0 0 0	1 0 0 0
1 0 0 0	1 0 0 1	1 0 0 1
1 0 0 1	1 0 1 0	1 0 1 0
1 0 1 0	1 0 1 1	1 0 1 1
1 0 1 1	1 1 0 0	1 1 0 0
1 1 0 0	1 1 0 1	1 1 0 1
1 1 0 1	0 0 0 0	0 0 0 0

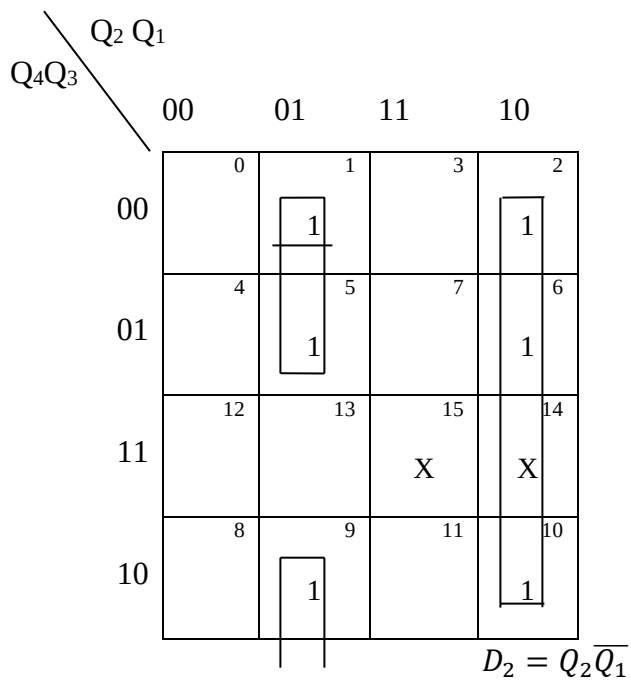
Step 4: The minimal expressions for D_4, D_3, D_2, and D_1 are obtained using K-maps as shown below.

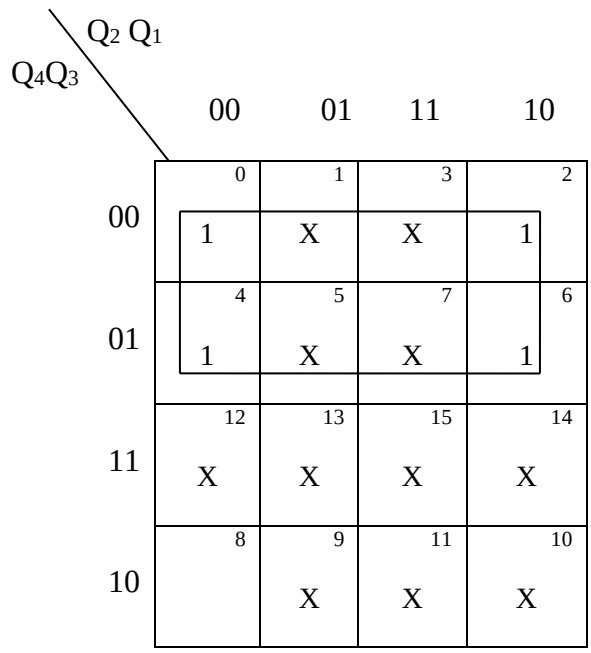




$D_4 = Q_4\overline{Q_3} + Q_4\overline{Q_1} + Q_3Q_2Q_1$

$D_3 = \overline{Q_4}Q_3Q_2 + \overline{Q_3}Q_2Q_1 + Q_3\overline{Q_1}$





$D_2 = Q_2\overline{Q_1} + \overline{Q_3}\,\overline{Q_2}Q_1 + \overline{Q_4}\,\overline{Q_2}Q_1$

$D_1 = \overline{Q_1}$

K-maps

The design equations for a mod-14 synchronous counter using D flip-flops are

$D_4 = Q_4\overline{Q_3} + Q_4\overline{Q_1} + Q_3Q_2Q_1$ $\qquad$ $D_3 = \overline{Q_4}Q_3\overline{Q_2} + \overline{Q_3}Q_2Q_1 + Q_3\overline{Q_1}$

$D_2 = Q_2\overline{Q_1} + \overline{Q_3}\,\overline{Q_2}Q_1 + \overline{Q_4}\,\overline{Q_2}Q_1$ $\qquad$ $D_1 = \overline{Q_1}$

A logic diagram can be drawn based on these expressions.

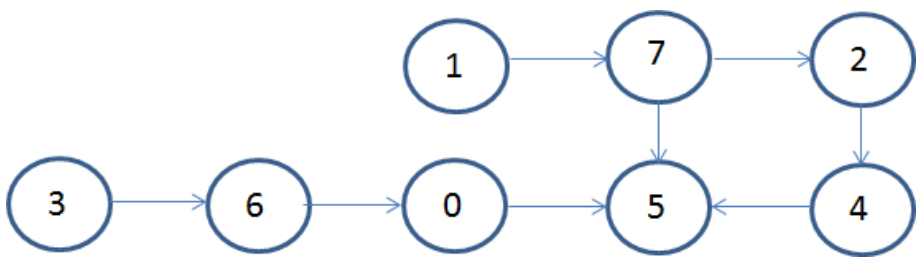
2 Design a J-K counter that goes through states 2, 4, 5, 7, 2, 4,…… Is the counter self- starting? Modify the circuit such that whenever it goes to any invalid sate it comes back to state 2.

Solution:

The J-K counter has to go through the states 2, 4, 5, 7, 2. . .

Step 1: The counter has only 4 stable states, but the highest count is 7. So it requires 3 flip flops. Three flip-flops can have 8 states. The remaining states 0, 1, 3, 6(i.e., 000, 001, 011, 110) are don't cares. The entries for excitations corresponding to these invalid states are don't cares.

Step 2: The state diagram of the 2, 4, 5, 7, 2 . . . counter is shown below.



State diagram

Step 3: J-K flip-flops are selected and the excitation table of the counter using J-K flip-flops is shown below.

Excitation table

PS	**NS**	**Required Excitations**
Q_3 Q_2 Q_1	**Q_3 Q_2 Q_1**	**J_3 K_3 J_2 K_2 J_1 K_1**
0 1 0	1 0 0	1 X X 1 0 X
1 0 0	1 0 1	X 0 0 X 1 X
1 0 1	1 1 1	X 0 1 X X 0
1 1 1	0 1 0	X 1 X 0 X 1

Step 4: From the excitation table $J_3 = 1$. The K-maps for excitations K_3, J_2, K_2, J_1 and K_1 their minimization and the minimal expressions obtained from them are shown below.

Q_3 \ Q_2Q_1	00	01	11	10
0	X (0)	X (1)	X (3)	X (2)
1	(4)	(5)	1 (7)	X (6)

$K_3 = Q_2$

Q_3 \ Q_2Q_1	00	01	11	10
0	X (0)	X (1)	X (3)	X (2)
1	(4)	1 (5)	X (7)	X (6)

$J_2 = Q_1$

Q_3 \ Q_2Q_1	00	01	11	10
0	X (0)	X (1)	X (3)	1 (2)
1	X (4)	X (5)	(7)	X (6)

$K_2 = \overline{Q_1}$

Q_3 \ Q_2Q_1	00	01	11	10
0	X (0)	X (1)	X (3)	(2)
1	1 (4)	X (5)	X (7)	X (6)

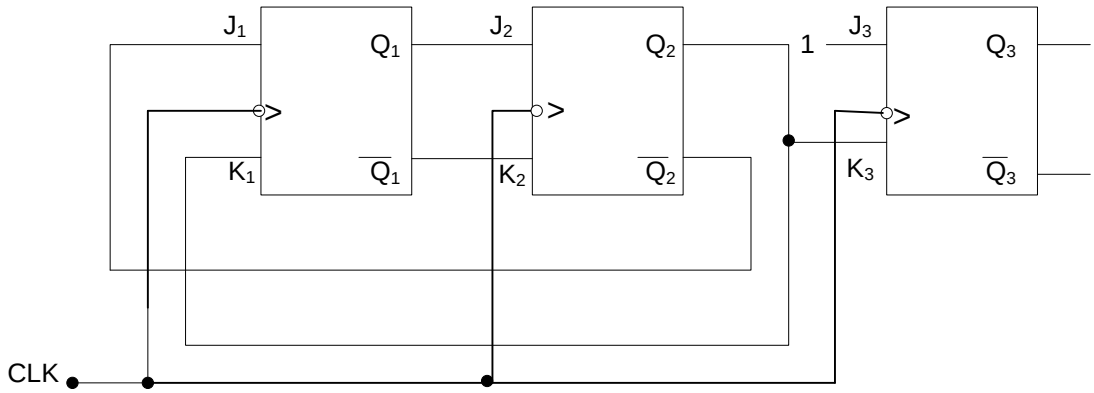
$J_1 = \overline{Q_2}$

Q_3 \ Q_2Q_1	00	01	11	10
0	X (0)	X (1)	X (3)	X (2)
1	X (4)	(5)	X (7)	X (6)

$K_1 = Q_2$

K- maps

Step 5: The logic diagram for the counter based on those minimal expressions is drawn as shown below.



Logic Diagram

Test for lock out:

Table to check for lock out

PS	Present Inputs	NS
$Q_3\ Q_2\ Q_1$	$J_3\ K_3$ $J_2\ K_2$ $J_1\ K_1$	$Q_3\ Q_2\ Q_1$
0 0 0	1 0 0 1 1 0	1 0 1
0 0 1	1 0 1 0 1 0	1 1 1
0 1 1	1 1 1 0 0 1	1 1 0
1 1 0	1 1 0 1 0 1	0 0 0

The counter is self-starting because from an invalid state it goes to a valid state after one or more clock pulses. To see that the counter goes to state 2(010) whenever it enters any of the invalid states, we obtain an expression for reset R as shown in table below and modify the circuit accordingly such that the counter goes to “010” after the next clock pulse.

Table for reset pulse

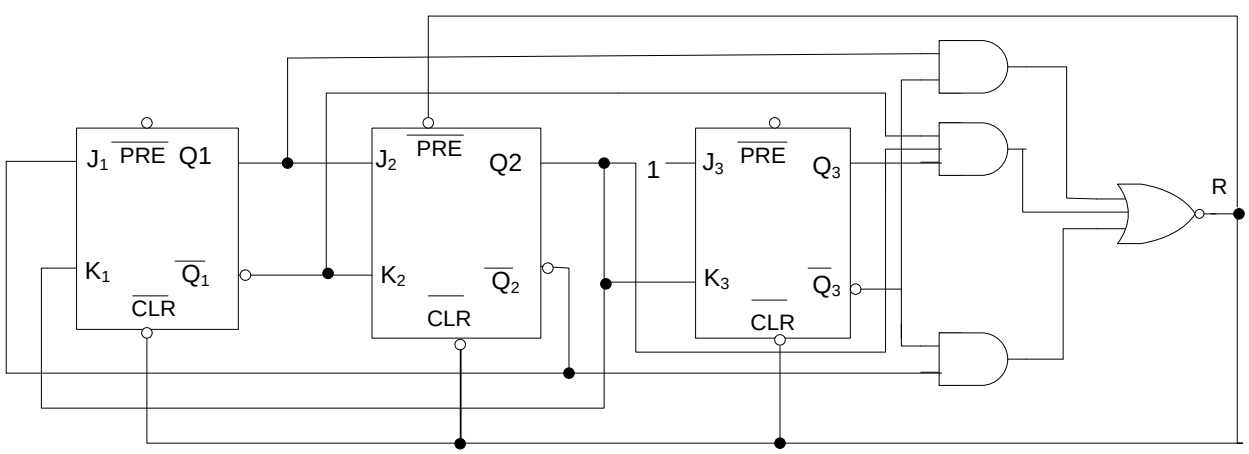
PS	Reset
Q_3 Q_2 Q_1	**R**
0 0 0	1
0 0 1	1
0 1 1	1
1 1 0	1

Q_3 \ $Q_2 Q_1$	00	01	11	10
0	0: 1	1: 1	3: 1	2
1	4	5	7	6: 1

$$R = \overline{Q_3}\,\overline{Q_2} + \overline{Q_3}\,Q_1 + Q_3 Q_2 \overline{Q_1}$$

K-map

The counter modified to go to state 2 whenever it enters any of the invalid states is shown below.



Modified logic diagram

3 Design the following types of counters:

(a) A type D counter that goes through states 0, 2, 4, 6, 0.... The undesired states must always go to a 0 on the next clock pulse.

(b) A type T counter that goes through states 6, 3, 7, 8, 2, 9, 1, 12, 14, 0, 6, 3 . . .

Is the counter self starting?

(c) A type T counter that goes through states 0, 5, 4, 2, 0….. Is the counter self starting?

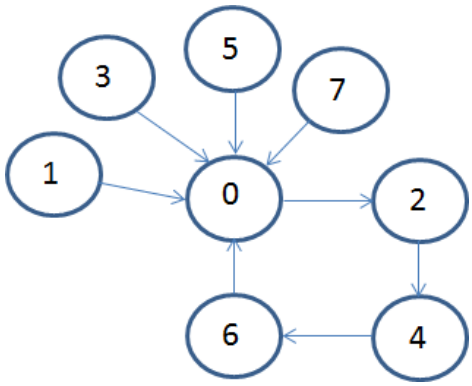
(d) A type T counter that must go through states 0, 2, 4, 6, 0 . . . if the control line is HIGH, and through states 0, 4, 2, 7, 0 . . . if the control line is LOW.

Solution:

(a) Type D counter that goes through states 0, 2, 4, 6, 0.... The undesired states must always go to a '0' on the next clock pulse.

Step 1: 3 flip-flops are required since the maximum count is 6(110). There are only 4 valid states 0, 2, 4, 6. So there are 4 invalid states (001, 011, 101, and 111). The counter has to go to '0' from each one of the undesired states.

Step 2: The state diagram is shown below.



State diagram

Step 3: The excitation table of the counter using D flip-flops is shown below.

Excitation table

PS	NS	Required Excitations
$Q_3\ Q_2\ Q_1$	$Q_3\ Q_2\ Q_1$	$D_3\quad D_2\quad D_1$
0 0 0	0 1 0	0 1 0
0 0 1	0 0 0	0 0 0
0 1 0	1 0 0	1 0 0
0 1 1	0 0 0	0 0 0
1 0 0	1 1 0	1 1 0
1 0 1	0 0 0	0 0 0
1 1 0	0 0 0	0 0 0
1 1 1	0 0 0	0 0 0

Step 4: The minimal expressions for D_3, D_2, and D_1 based on the excitation table are

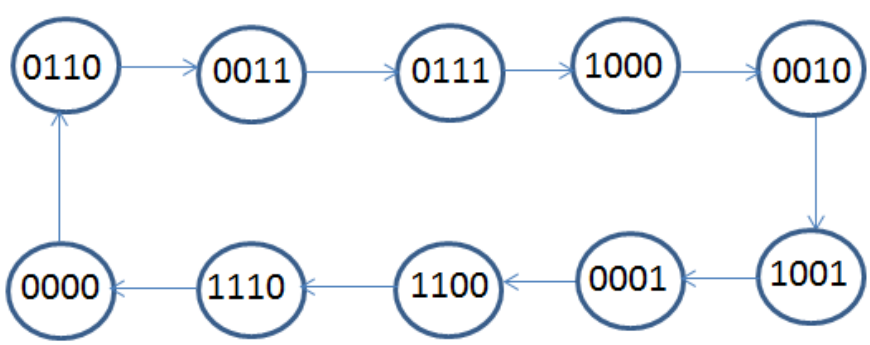
$D_3 = \overline{Q_3}Q_2\overline{Q_1} + Q_3\overline{Q_2}\,\overline{Q_1}$ $\qquad D_2 = \overline{Q_3}\,\overline{Q_2}\,\overline{Q_1} + Q_3\overline{Q_2}\,\overline{Q_1} = \overline{Q}_2\,\overline{Q}_1$ $\qquad D_1 = 0$

Step 5: A logic diagram can be drawn using the excitations of step 4 using 3 D flip-flops.

(b) A type T counter that goes through states 6, 3, 7, 8, 2, 9, 1, 12, 14, 0, 6, 3 . . .

Step 1: The maximum count is 14(1110). So it is a 4-bit counter. It can have 16 states. There are 10 valid states so the remaining 6 states 4, 5, 10, 11, 13, and 15(i.e., 0100, 0101, 1010, 1011, 1101, and 1111) are invalid. The entries for excitations corresponding to these states are don't cares.

Step 2: The state diagram of the counter is shown below.



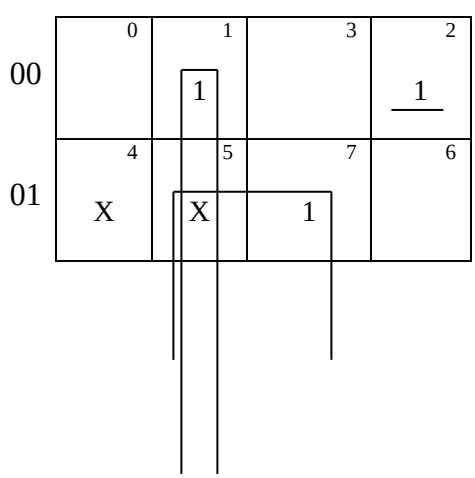
State diagram

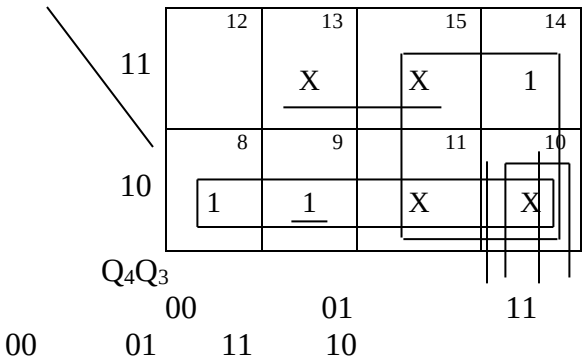
Step 3: The excitation table of the counter using T flip-flops is shown below.

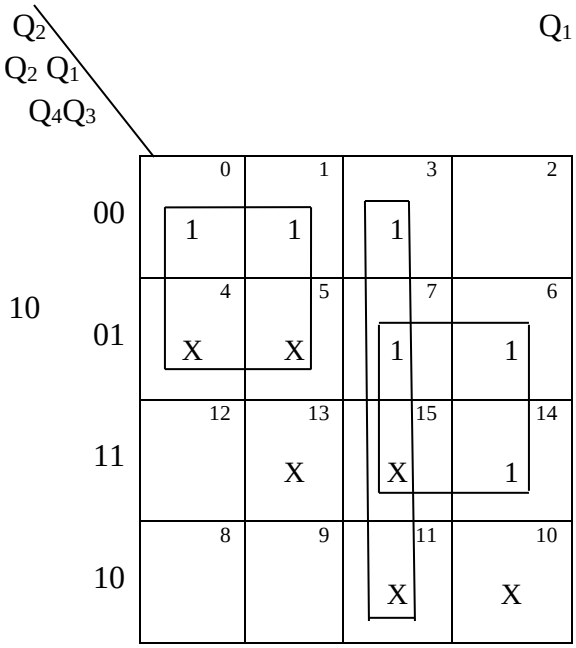
Excitation table

PS	**NS**	**Required Excitations**
Q_4 Q_3 Q_2 Q_1	**Q_4 Q_3 Q_2 Q_1**	**T_4 T_3 T_2 T_1**
0 1 1 0	0 0 1 1	0 1 0 1
0 0 1 1	0 1 1 1	0 1 0 0
0 1 1 1	1 0 0 0	1 1 1 1
1 0 0 0	0 0 1 0	1 0 1 0
0 0 1 0	1 0 0 1	1 0 1 1
1 0 0 1	0 0 0 1	1 0 0 0
0 0 0 1	1 1 0 0	1 1 0 1
1 1 0 0	1 1 1 0	0 0 1 1
1 1 1 0	0 0 0 0	1 1 1 0
0 0 0 0	0 1 1 0	0 1 1 0

Step 4: The minimal expressions for excitations T_4, T_3, T_2, and T_1 can be obtained using K-maps as shown below

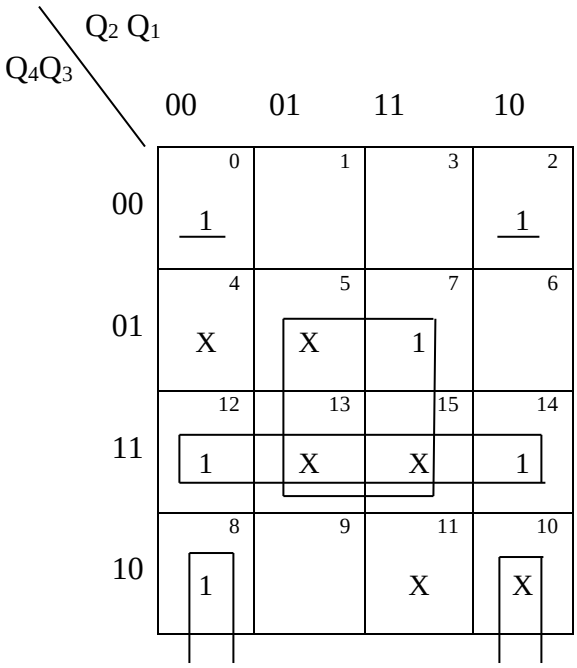


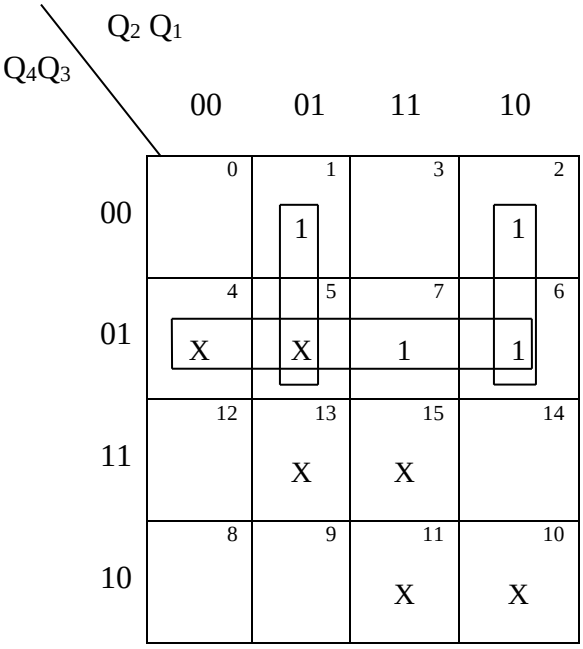




$$T_4 = Q_4\overline{Q_3} + \overline{Q_2}\,Q_1 + Q_3Q_1 + Q_4Q_2 + \overline{Q_3}Q_2\overline{Q_1}$$

$$T_3 = Q_3Q_2 + Q_2Q_1 + \overline{Q_4}\,\overline{Q_2}$$





$$T_2 = \overline{Q_3}\,\overline{Q_1} + Q_4Q_3 + Q_3Q_1 \qquad T_1 = \overline{Q_4}Q_3 + \overline{Q_4}\,\overline{Q_2}Q_1 + \overline{Q_4}\,Q_2\,\overline{Q_1}$$

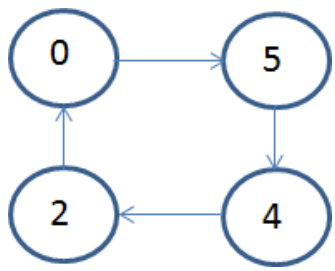
K-maps

Step 5: A logic diagram using 3 T flip-flops can be drawn using the equations obtained for T_4, T_3, T_2, and T_1.

(c) A type T counter that goes through states 0, 5, 4, 2, and 0.

Step 1: The highest count is 5(101). So 3 flip-flops are required. The counter has only four valid states 0, 5, 4, and 2. The remaining four states 1, 3, 6, and 7 are invalid. The entries for excitations corresponding to these states are don't cares.

Step 2: The state diagram is shown below.



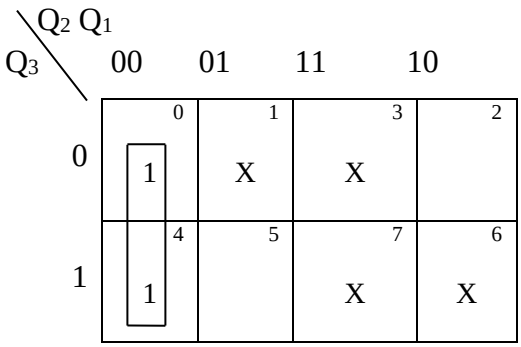
State diagram

Step 3: The excitation table for counter using T flip-flops is shown below.

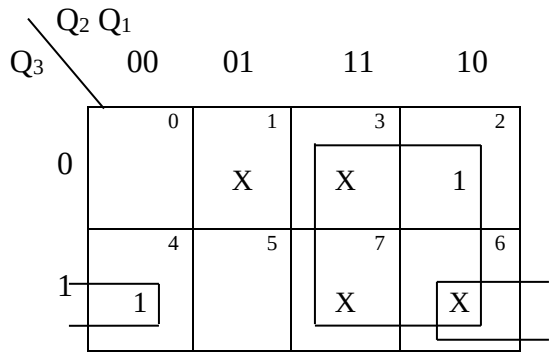
Excitation table

PS	**NS**	**Required Excitations**
Q_3 Q_2 Q_1	**Q_3 Q_2 Q_1**	**T_3 T_2 T_1**
0 0 0	1 0 1	1 0 1
1 0 1	1 0 0	0 0 1
1 0 0	0 1 0	1 1 0
0 1 0	0 0 0	0 1 0

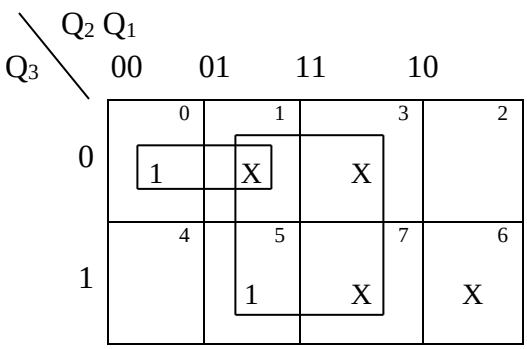
Step 4: The K-maps, their minimization and the minimal expressions for T_3, T_2, and T_1 obtained from them are shown below.



$T_3 = \overline{Q_3}\,\overline{Q_1}$



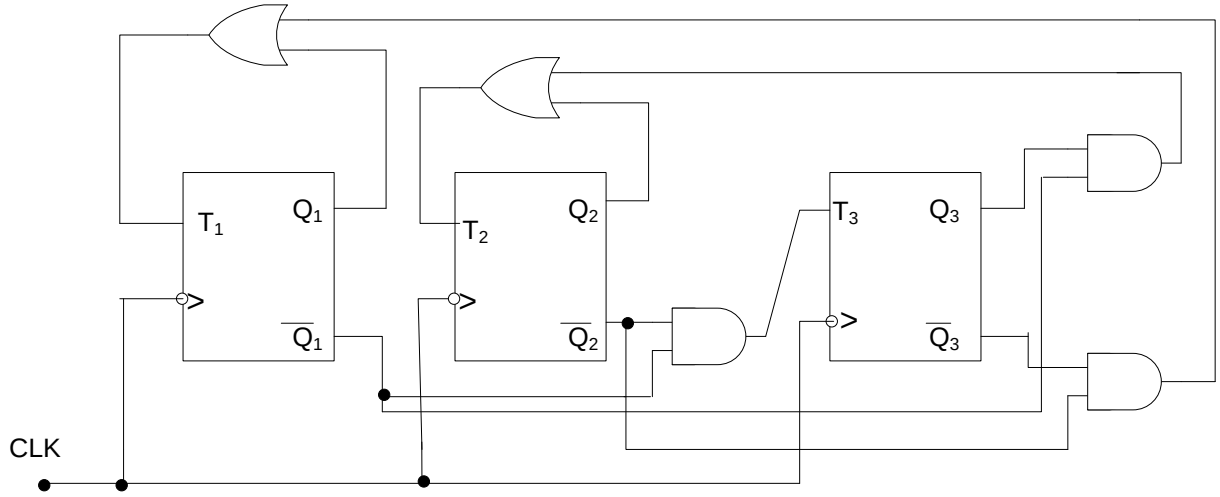
$T_2 = Q_3\overline{Q_1} + Q_2$



$T_1 = \overline{Q_3}\,\overline{Q_2} + Q_1$

K-maps

Step 5: The logic diagram based on the minimal expressions is shown in figure below.



Logic diagram

Step 6: The next state entries for the counter as per the logic diagram based on present state entries are shown in table to check for lockout shown below.

Table to check for lockout

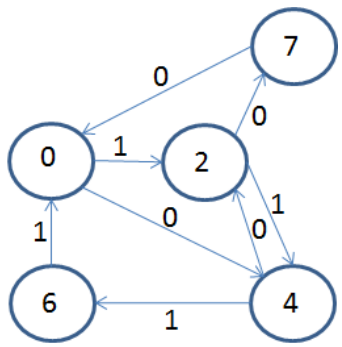
PS	**Present Inputs**	**NS**
Q_3 Q_2 Q_1	**T_3 T_2 T_1**	**Q_3 Q_2 Q_1**
0 0 1	0 0 1	0 0 0
0 1 1	0 1 1	0 0 0
1 1 0	0 1 0	1 0 0
1 1 1	0 1 1	1 0 0

The counter is self-starting as it goes to valid state from all invalid states after the next clock pulse.

(d) Type T counter that must go through states 0, 2, 4, 6, 0 . . . if the control line is high and through states 0, 4, 2, 7, 0 . . . if the control line is low.

Step 1: The maximum count is 7. So the counter requires 3 FFs. The counter has five states 0, 2, 4, 6, and 7. The remaining 3 states 1, 3, 5 are invalid. The entries for excitations corresponding to these invalid states are don't cares.

Step 2: The state diagram of the counter is shown below.



State diagram

Step 3: The excitation table of the counter using T FFs is shown below.

Excitation table

PS		NS	Required Excitations
$Q_3\ Q_2\ Q_1$	M	$Q_3\ Q_2\ Q_1$	T_3 T_2 T_1
0 0 0	0	1 0 0	1 0 0
0 0 0	1	0 1 0	0 1 0
0 1 0	0	1 1 1	1 0 1
0 1 0	1	1 0 0	1 1 0
1 0 0	0	0 1 0	1 1 0
1 0 0	1	1 1 0	0 1 0
1 1 0	0	-------	--------
1 1 0	1	0 0 0	1 1 0
1 1 1	0	0 0 0	1 1 1
1 1 1	1	-------	--------

The don't cares are minterms 2, 3, 6, 7, 10, 11, 12, and 15. The expressions for excitations are

$T_3 = \sum m\ (0, 4, 5, 8, 13, 14) + d\ (2, 3, 6, 7, 10, 11, 12, 15)$

$T_2 = \sum m\ (1, 5, 8, 9, 13, 14) + d\ (2, 3, 6, 7, 10, 11, 12, 15)$

$T_1 = \sum m\ (4, 14) + d\ (2, 3, 6, 7, 10, 11, 12, 15)$.

Step 4: The minimal expressions for excitations T_3, T_2, and T_1 are obtained using K-maps as shown below.

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0: 1	1:	3: X	2: X
01	4: 1	5: 1	7: X	6: X
11	12: X	13: 1	15: X	14: 1
10	8: 1	9:	11: X	10: X

$$T_3 = Q_2 + \bar{M}$$

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0:	1: 1	3: X	2: X
01	4:	5: 1	7: X	6: X
11	12: X	13: 1	15: X	14: 1
10	8: 1	9: 1	11: X	10: X

$$T_2 = Q_3 + M$$

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0:	1:	3: X	2: X
01	4: 1	5:	7: X	6: X
11	12: X	13:	15: X	14: 1
10	8:	9:	11: X	10: X

$$T_1 = Q_2\bar{M}$$

K-maps

Step 5: A logic diagram using T FFs can be drawn based on the equations for T_3, T_2, and T_1.

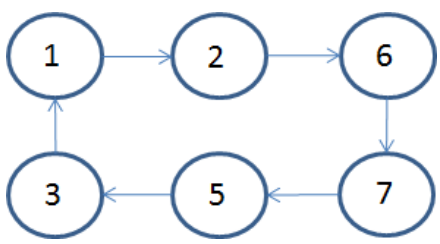
4. Design a direct logic circuit to generate the sequences 100111 and 011101 simultaneously.

Solution:

To design a direct logic circuit to generate the sequences 100111 and 011101 simultaneously the state assignment is done as shown below. There must be 6 unique states so at least 3 FFs are required. In this case 6 unique states are obtained using 3 FFs. Let us design the sequence generator using D FFs.

State assignment

States	**Q_3 Q_2 Q_1**
1	0 0 1
2	0 1 0
6	1 1 0
7	1 1 1
5	1 0 1
3	0 1 1



State diagram

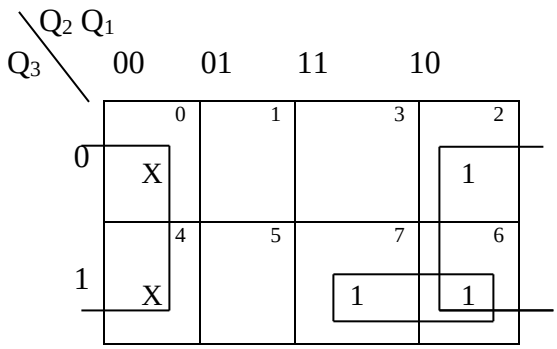
The excitation table when D FFs are used is shown below. States '0' and '4' are invalid. The entries for excitations corresponding to those states are invalid.

Excitation table

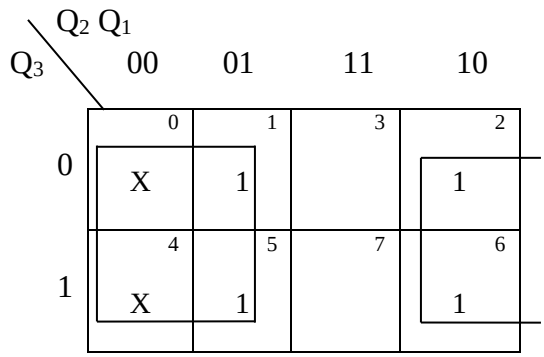
PS	**NS**	**Required Excitations**
Q_3 Q_2 Q_1	**Q_3 Q_2 Q_1**	**D_3 D_2 D_1**
0 0 1	0 1 0	0 1 0
0 1 0	1 1 0	1 1 0
1 1 0	1 1 1	1 1 1

1 1 1	1 0 1	1 0 1
1 0 1	0 1 1	0 1 1
0 1 1	0 0 1	0 0 1

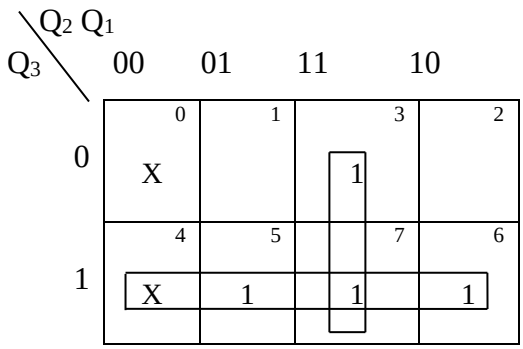
The minimal expressions for D_3, D_2, and D_1 based on the above table are obtained using K-maps shown below



$D_3 = \overline{Q_1} + Q_3Q_2$



$D_2 = \overline{Q_1} + \overline{Q_2}$



$D_1 = Q_3 + Q_2Q_1$

K-map

5. Generate the pulse train 100110 using indirect logic.

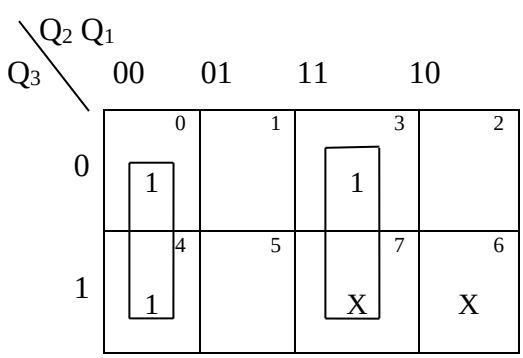
Solution:

To generate the pulse train 100110 using indirect logic write the sequence vertically down under the column heading output (f) in the truth table. Since the sequence is 6 bits long, we need 6 unique states to generate the above pulse train. So any mod-6 counter can be used. For simplicity

we use a ripple counter. It goes through states 0, 1, 2, 3, 4, 5, 0 . . . States 6 and 7 are invalid, so the corresponding outputs are don't cares. The K-map for the output f in terms of the outputs of the FFs, their minimization and the minimal expression obtained from it are shown in figure.

Truth table

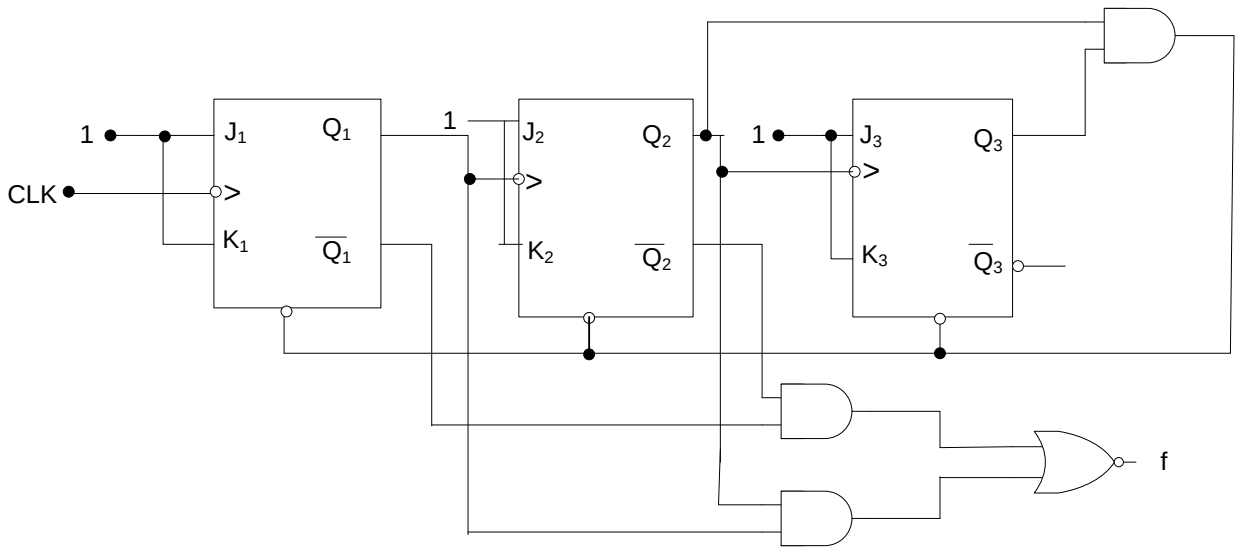
States	**Q_3 Q_2 Q_1**	**f**
0	0 0 0	1
1	0 0 1	0
2	0 1 0	0
3	0 1 1	1
4	1 0 0	1
5	1 0 1	0
6	1 1 0	X
7	1 1 1	X



$$f = \overline{Q_2}\,\overline{Q_1} + Q_2 Q_1$$

K-map

A 3-bit ripple counter using J-K FFs is used. Once it enters state 110 it should be reset to "000". The logic diagram is shown below.



Logic diagram

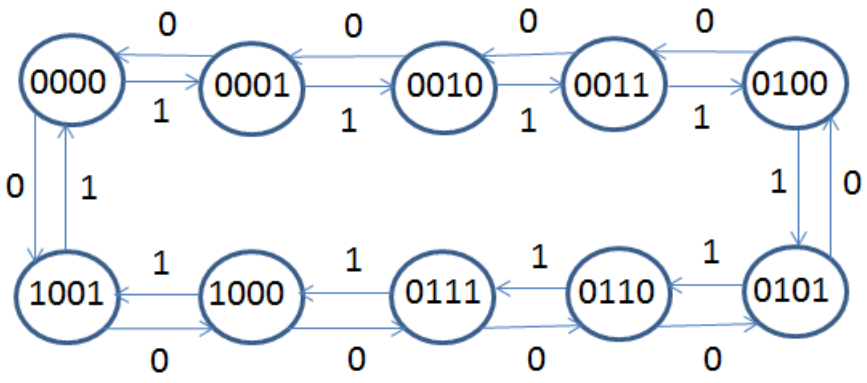
6 Design a BCD up/ down counter using S-R FFs

Solution:

Design of a synchronous BCD up/down counter using S-R FFs.

Step 1: BCD counter is a 4-bit counter. For up/down counting a mode signal is required. So 4 FFs are required but the minimal expressions for excitations are obtained using 5 variable K- maps.

Step 2: The state diagram of the BCD up/down counter is shown below.



State diagram

Step 3: The excitation table of the BCD up/down counter using S-R FFs is shown below.

Excitation table

PS	Mode	NS	Required Excitations
Q_4 Q_3 Q_2 Q_1	M	Q_4 Q_3 Q_2 Q_1	S_4 R_4 S_3 R_3 S_2 R_2 S_1 R_1
0 0 0 0	0	1 0 0 1	1 0 0 X 0 X 1 0
0 0 0 0	1	0 0 0 1	0 X 0 X 0 X 1 0
0 0 0 1	0	0 0 0 0	0 X 0 X 0 X 0 1
0 0 0 1	1	0 0 1 0	0 X 0 X 1 0 0 1
0 0 1 0	0	0 0 0 1	0 X 0 X 0 1 1 0
0 0 1 0	1	0 0 1 1	0 X 0 X X0 1 0
0 0 1 1	0	0 0 1 0	0 X 0 X X0 0 1
0 0 1 1	1	0 1 0 0	0 X 1 0 0 1 0 1
0 1 0 0	0	0 0 1 1	0 X 0 1 1 0 1 0
0 1 0 0	1	0 1 0 1	0 X X 0 0 X 1 0
0 1 0 1	0	0 1 0 0	0 X X 0 0 X 0 1
0 1 0 1	1	0 1 1 0	0 X X 0 1 0 0 1
0 1 1 0	0	0 1 0 1	0 X X 0 0 1 1 0
0 1 1 0	1	0 1 1 1	0 X X 0 X 0 1 0
0 1 1 1	0	0 1 1 0	0 X X 0 X 0 0 1
0 1 1 1	1	1 0 0 0	1 0 0 1 0 1 0 1
1 0 0 0	0	0 1 1 1	0 1 1 0 1 0 1 0
1 0 0 0	1	1 0 0 1	X 0 0 X 0 X 1 0
1 0 0 1	0	1 0 0 0	X 0 0 X 0 X 0 1
1 0 0 1	1	0 0 0 0	0 1 0 X 0 X 0 1

5 variables, (Q_4, Q_3, Q_2, Q_1, and M) can have 32 combinations. The 20 combinations shown in the excitation table are valid. The remaining 12 (20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, and 31) are invalid. The entries for excitations corresponding to these combinations are don't cares. Therefore we have from the excitation table

$S_4 = \sum m\ (0, 15) + d\ (8, 9, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31)$

$R_4 = \sum m\ (16, 19) + d\ (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31)$

$S_3 = \sum m\ (7, 16) + d\ (9, 10, 11, 12, 13, 14, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31)$

$R_3 = \sum m\ (8, 15) + d\ (0, 1, 2, 3, 4, 5, 6, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31)$

$S_2 = \sum m\ (3, 8, 11, 16) + d\ (5, 6, 13, 14, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31)$

$R_2 = \sum m\ (4, 7, 12, 15) + d\ (0, 1, 2, 9, 10, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31)$

$S_1 = \sum m\ (0, 1, 4, 5, 8, 9, 12, 13, 16, 17) + d\ (20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31)$

$R_1 = \sum m\ (2, 3, 6, 7, 10, 11, 14, 15, 18, 19) + d\ (20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31)$

The minimal expressions for the above excitations can be obtained using K-maps.

Q_4

Q_4Q_3 \ Q_2Q_1	00	01	10	11
00	0: 1	1	3	2
01	4	5	7	6
11	12	13	15: 1	14
10	8	9	11	10

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	16	17: X	19	18: X
01	20: d	21: d	23: d	22: d
11	28: d	29: d	31: d	30: d
10	24: d	25: d	27: d	26: d

$$S_4 = \overline{Q_4}\,\overline{Q_3}\,\overline{Q_2}\,\overline{Q_1}\,\overline{M}$$

Similarly from the K-maps we get

$$R_4 = Q_4\overline{Q_1}\,\overline{M} + Q_4Q_1M$$

$S_3 = \overline{Q_3}\,Q_2\,Q_1\,M + Q_4\overline{Q_1}\,\overline{M}$ $\qquad$ $R_3 = Q_3Q_2Q_1M + Q_3\,\overline{Q_2}\,\overline{Q_1}\,\overline{M}$

$S_2 = \overline{Q_4}\,\overline{Q_2}\,Q_1M + Q_4\overline{Q_1}\,\overline{M} + Q_3\,\overline{Q_2}\,\overline{Q_1}\,\overline{M}$ $\qquad$ $R_2 = Q_2Q_1M + Q_2\overline{Q_1}\,\overline{M}$

$S_1 = \overline{Q_1}$ $\qquad$ $R_1 = Q_1$

Based on the above expressions a logic diagram can be drawn for a BCD counter using S-R FFs.

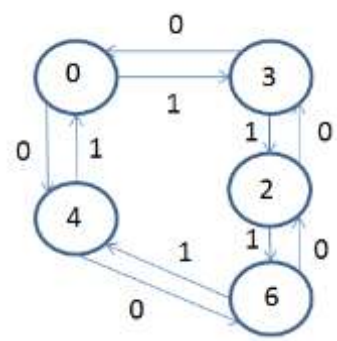
7 Design an up/down counter using D FFs to count 0, 3, 2, 6, 4, 0….

Solution:

Design of an up/down counter using D FFs to count 0, 3, 2, 6, 4, 0…

Step 1: The maximum count is 6. So 3 FFs are required. 3 FFs can have 8 states. 5 are valid. So the remaining 3(1, 5, and 7) are invalid. It is an up/down counter, so a mode signal is required. So there are 4 variables Q_3, Q_2, Q_1, and M, i.e., the output of 3 FFs and mode signal. Out of 16 (2^4) combinations 10 are valid and 6 are invalid. The entries for excitations corresponding to these combinations are don't cares.

Step 2: The state diagram of the 0, 3, 2, 6, 4, 0… up/down counter is shown below.



State diagram

Step 3: The excitation table of the counter using D FFs is shown below.

Excitation table

PS	Mode	NS	Required Excitations
Q_3 Q_2 Q_1	**M**	**Q_3 Q_2 Q_1**	**D_3 D_2 D_1**

0 0 0	0	1 0 0	1 0 0
0 0 0	1	0 1 1	0 1 1
0 1 1	0	0 0 0	0 0 0
0 1 1	1	0 1 0	0 1 0
0 1 0	0	0 1 1	0 1 1
0 1 0	1	1 1 0	1 1 0
1 1 0	0	0 1 0	0 1 0
1 1 0	1	1 0 0	1 0 0
1 0 0	0	1 1 0	1 1 0
1 0 0	1	0 0 0	0 0 0

The valid combinations are m (0, 1, 6, 7, 4, 5, 12, 13, 8, and 9). The invalid combinations are d (2, 3, 10, 11, 14, and 15). From the excitation table we have

$D_3 = \sum m (0, 5, 8, 13) + d (2, 3, 10, 11, 14, 15)$

$D_2 = \sum m (1, 4, 5, 7, 8, 12) + d (2, 3, 10, 11, 14, 15)$

$D_1 = \sum m (1, 4) + d (2, 3, 10, 11, 14, 15)$

Step 4: The K-maps for excitations and the minimal expressions obtained from them are shown below.

Q_4Q_3 \ Q_2Q_1	00	01	10	11
00	0: 1	1	3: X	2: X
01	4	5: 1	7	6
11	12	13: 1	15: X	14: X
10	8: 1	9	11: X	10: X

$$D_3 = \overline{Q_2}\overline{M} + Q_2\,\overline{Q_1}M$$

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0	1: 1	3: X	2: X
01	4: 1	5: 1	7: 1	6
11	12: 1	13	15: X	14: X
10	8: 1	9	11: X	10: X

$$D_2 = Q_3\overline{M} + \overline{Q_3}M + Q_2\,\overline{Q_1}\,\overline{M}$$

Q_4Q_3 \ Q_2Q_1	00	01	11	10
00	0	1: 1	3: X	2: X
01	4: 1	5	7	6
11	12	13	15: X	14: X
10	8	9	11: X	10: X

$$D_1 = \overline{Q_3}\,\overline{Q_2}M + \overline{Q_3}Q_2\,\overline{Q_1}\,\overline{M}$$

K-maps